Continuous Convergence and Curried Functional Spaces

1. Preliminary definitions

Definition 1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 1.2. A filter F on X is called an *ultrafilter* if is not properly contained in any other filter. A standard application of Zorn's lemma shows that every filter in contained in at least one ultrafilter [1]. Moreover, for any filter $F \in \mathcal{F}(X)$ we have

$$F = \bigcap\{G \mid G \text{ is an ultrafilter and } F \subset G\}.$$

Definition 1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \to x$, if it contains all open neighborhoods of x.

Remark 1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter

  $$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$

  converges to x. This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.
- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \varnothing$ for all $A \in F$.
- A function $f : X \to Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \to x \in X$ implies $f(F) \to f(x) \in Y$.

Definition 1.4. Let X, Y be topological spaces. By $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y.

Bibliography

[1] N. Bourbaki, *General Topology, Part I*. 1966.